

Service recovery planning in OpenStack is a critical administrative task focused on ensuring **service continuity**, minimizing downtime, and protecting data across a complex, multi-component infrastructure 1, 2.

Fault Tolerance Strategies

OpenStack achieves fault tolerance primarily through physical and logical segregation of resources:

- **Availability Zones (AZ):** Compute nodes are grouped into **fault domains** based on physical location or power sources 3, 4. This ensures that if one zone fails due to a natural force or hardware issue, instances in other zones remain unaffected 1, 3.
- **Symmetric Clustering:** In this distributed approach, all nodes are peers with equal capabilities 5. This eliminates the **single point of failure** inherent in asymmetric (centralized) models, as tasks are distributed across the cluster 5, 6.
- **Nova-Scheduler Logic:** The scheduler respects placement preferences for **redundancy**, strategically distributing VMs to ensure they do not all reside on a single failing node 7, 8.

Backup Mechanisms

The sources emphasize that OpenStack does not provide a single "special tool" for backups; instead, it is a collection of components that must each be backed up individually 2.

- **Component Mapping:** Operators must map every infrastructure component and prepare an **easy, efficient, and auto-recovery enabled** strategy for each 2.
- **Target Data:** Regular backups should include **virtual machine data** and **cloud configurations** to ensure full recovery 9.
- **Storage Backends: Object storage (Swift)** is highlighted as an ideal alternative for storing unstructured data like backups and archives 10, 11.

High Availability (HA) Planning

High availability planning ensures that services remain accessible even during hardware or software failures:

- **Clustering and Redundancy:** Implementing **clustering strategies** ensures that if one node fails, the service can continue running on another 9.
- **Load Balancing:** Segregation strategies help maintain **low latency** in API services, which is essential for responsive HA environments 1.
- **Placement Groups:** Administrators use these in the nova-scheduler to mandate that highly available VMs are placed in different physical locations 7, 8.

Failover Techniques

Failover mechanisms allow for the rapid restoration of instances after a crash:

- **External Shared Storage:** Storing instance disks on an external pool rather than local disks is a primary failover technique 12. If a compute node fails, instances can be **recovered easily** because their data resides safely on the shared storage pool 12.

- **Automated Action Handling:** The **Senlin service** uses worker threads to monitor the database for "READY" actions, allowing for asynchronous recovery operations when an engine receiver triggers a failover event 13, 14.

Disaster Recovery (DR) Considerations

A disaster recovery plan is mandatory for production environments to handle major outages 9, 15.

- **Risk Mitigation:** Starting production without a DR background is considered **highly risky**, and immediate planning is required 16.
- **Infrastructure Recovery:** A DR plan must outline how to recover the **entire cloud infrastructure**, not just individual VMs, in the event of a catastrophic failure 9, 15.

Exam-Oriented Explanation

- **The "No Single Tool" Rule:** OpenStack requires a **component-by-component backup strategy** because it is a modular system 2.
- **Availability Zones vs. Clustering:** **AZs** provide physical fault tolerance (hardware level), while **Symmetric Clustering** provides architectural fault tolerance (service level) 3, 5.
- **Recovery Requirement:** Successful recovery relies on **external shared storage** to decouple instance data from the physical health of a single compute node 12.
- **Telemetry's Role:** **Telemetry services** (Ceilometer) provide the historical event logs and resource data necessary for **troubleshooting** and identifying the root cause of service failures 17, 18.
- **Key Services Involved:** **Keystone** (re-authentication after recovery), **Cinder** (re-attaching block storage), and **Neutron** (re-establishing virtual networking connectivity) 11, 19, 20.